NAN PROVINCE THAILAND



Rice Farmer Field School

Silaphet Sub-district · Pua District · Nan Province

Learning rice, health and Tai Lue life — in the paddies, alongside local farmers

Background

The traditional farming system in Silaphet sub-district involved the cultivation of paddy rice followed by a variety of vegetable crops after the rice harvest, grown for household consumption. Between 1987 and 1992, farmers in the area moved from subsistence into commercial crop cultivation, particularly chili, a shift driven by middlemen who promoted chili cultivation and offered good cash income. On the other hand, the shift from rice transplanting to direct seeding as a labor-saving practice generalized in the area, as in other provinces of the country engaged in a general process of agricultural intensification.

The new cultivation practices led farmers to invest in more chemicals, especially herbicide and insecticides in former "organic by default" rice fields. Problems of decline in soil fertility, yields and crop failures were brought to the attention of the sub-district (tambon) agricultural office (TAO), and experts suggested substituting chili with other crops: peanut, soybean, cucumber, Chinese cabbage, pea and broccoli, to sustain yields and prevent disease. However, the lack of collective crop-selection decisions caused marketing problems, with some crops in over-supply and others in short supply.

Vegetable farmers therefore registered with the TAO to obtain advice on crop types and farm planning. Vegetable farmers in Silaphet established a 'safe vegetables producing group', supported by the sub-district hospital (which carried out random blood tests for chemical residues) and the local agricultural office (which advised on safe and appropriate chemical use). This collective brought the issues of safe food production and consumption to the attention of both farmers and consumers who wanted to reduce chemical use.



Origins of the Farmer Field School

Rice curriculum in Ban Thoung Khong

In 2011–12 (B.E. 2554–55), following a Ministry of Education directive for schools nationwide to develop curricula adapted to local situations and linked to community development, Mrs. Jongrak, then teaching natural sciences (biology) at Sattha Silaphet School, began building a curriculum around rice cultivation. She was in contact with the Joko community center, which worked with rice farmers on seed production, and together they developed activities in Ban Thoungkhong, Tambon Yom, Amphoe Thawangpha, leading to a farmer field school gathering 15 farming households. The approach echoed the King's philosophy of sufficiency economy, and Mrs. Jongrak joined the group to provide training support for her high-school students.

Together, farmers and students worked on seed saving, selection and demonstration of rice varieties (both sticky and non-glutinous), and on the study and conservation of local rice varieties with the support of the Hug Muang Nan foundation. Students often joined activities outside official class hours. The natural-science curriculum, built around rice as a study topic, was developed jointly with farmers, fully adapted to the local context, and evolved as farming practices in the area changed. It was taught directly in farmers' fields, such as Mrs. Phin's farm, with support from the Tambon authorities and the local health station in Silaphet, where Mrs. Suttana Pathamawat and her husband worked.

Mrs. Suttana's public health work

Mrs. Suttana and her husband, Mr. Sathian, were public health workers and academics at the sub-district health station, investigating rising rates of diabetes, hypertension, cancer and other non-communicable diseases. As part of this work and her MSc study, Mrs. Suttana tested villagers' blood cholinesterase levels, an enzyme test used to assess poisoning from the organophosphate and carbamate pesticides commonly used by farmers and others regularly exposed to agricultural chemicals.

Testing was carried out in every village across the sub-district, and it revealed very high exposure rates — the more they tested, the more cases they found. Wanting to understand the underlying cause, Mrs. Suttana launched a campaign against chemical use. At the time, she owned rice land with her siblings but did not farm it herself; others farmed it, using agrochemicals. Villagers pointed out the contradiction: "She campaigns against chemical use, yet her own land still uses chemicals"; so Mrs. Suttana decided to farm the land herself, without chemicals.

> ***"She campaigns against chemical use, yet her own land still uses chemicals."***
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> What villagers said was the turning point that led Mrs. Suttana to farm her land without chemicals.

Founding of the Rice Farmer Field School

Once Mrs. Suttana began farming the land herself, Mrs. Jongrak asked to use it as a learning space for school children to learn about farming, and the two women worked together to design a curriculum. Mrs. Jongrak's original native-rice curriculum, developed in Thung Khong village (Yom sub-district), fell outside Silaphet's jurisdiction, so the Sila Phet sub-district administrative organization (อบต., OBT) could not support the children studying there. In 2012, the class moved onto Mrs. Suttana's land in the neighboring village of Nakham in Silaphet sub-district, within the same jurisdiction as Mrs. Jongrak's Silaphet high school, and serving as a model for villagers alongside her own farming experiments. With OBT support, an organic market was created and the curriculum evolved to cover the full cycle of rice production, from nursery to harvest, processing and marketing, including training courses on weed and water management.





The curriculum grew from one classroom to two, then three, and the Farmer School was born, initially using roughly 5 rai of paddy land inherited by Mrs. Suttana together with her siblings, Mrs. Kanika and Mr. Sanith, who agreed to make part of the land available for student learning. Mrs. Kanika taught students the rice-farming process, and part of the harvest was used in a curriculum where students learned to process rice into sweets and to sell products through online and offline marketing. Guest accommodation followed in 2012–13 (B.E. 2555–56), built to generate income for the learning center; the "farm stay" accommodation was registered as a commercial business, while the Farmer Field School itself was registered as a learning center.

Training activities then expanded to safe vegetables grown after rice during the winter season: Mo4-grade students studied vegetable crops while Mo5-grade students worked on rice, including traditional variety conservation and labor-saving production techniques. Mrs. Suttana also traveled to Sing Buri province to obtain Riceberry rice seed, a low-sugar content variety, for testing. Students collected water and soil samples to measure chemical contamination and tested agricultural products and blood cholinesterase levels, to better understand the relationship between environmental contamination and human health. Farmers engaged in clean vegetable production to sell to the local hospital, certified by the hospital laboratory, which checked for contamination. Mrs. Suttana used the Farmer School as the subject of her MSc thesis, running training sessions, managing her public health work, and the school simultaneously. The school also began welcoming visiting learners and tourists, both to expose them to the project and to generate supplementary income.

In 2019 (B.E. 2562), Mrs. Suttana and Mr. Sathian resigned from their jobs to run the Farmer School full-time while keeping a health focus, after the rice season they grew vegetables on the same land, tested them, and sold them to Pua Hospital to support villagers' income. Asked how confident she was that homestay income could sustain their livelihood, Mrs. Suttana explained that the school did not require a large budget, so she felt able to manage, and believed tourism income would be enough to cover living costs.

TIMELINE SUMMARY

2011	2012–13	2013–18	2019
B.E. 2554	B.E. 2555–56	B.E. 2556–61	B.E. 2562
Farmer School founded (no proper school building yet, just a small room).	Guest rooms were built; began accepting tourists.	Farmer School in operation, while Mrs. Suttana still worked in public health.	Mrs. Suttana and Mr. Sathian resign from their jobs; income relies mainly on tourists.

Activities and Training Content

Rice cultivation curriculum

Before seedlings are grown, students learn to select seed varieties and test germination rates, then work alongside villagers doing the same tasks, learning directly from local farmers. Students practice meticulous rice-planting techniques, using straw thread to measure precise spacing and tying exact knots, learning the traditional planting process alongside academic principles such as how many tillers a single rice plant produces. This served as a model to show farmers that fewer seedlings can be used, both to save costs and to increase yield.

The curriculum was designed for children who attend regular school daily and often have extra tutoring on weekends, yet, despite being the children of rice-farming families, rarely had the chance to learn farming skills or how to use their own family's fields. A curriculum directly teaching these skills was designed to fill that gap, built to keep teaching costs low by relying on local people as instructors rather than paid outside teachers.





Vegetables are also grown after the rice season: chili, shallots, garlic, lettuce, Chinese cabbage, broccoli, using safe-vegetable cultivation principles. The project has encouraged the wider Silaphet community to grow safe vegetables too, through a community 'Health Charter' that sets quality standards. Produce is consumed within households, sold through middlemen at roadside stalls, and supplied to Pua Hospital, with quality inspected and certified as verified-safe by the Silaphet sub-district administration (OBT), giving consumers confidence in the safety of Silaphet produce.

Homestay development

OBT support for clean food and chemical-free agriculture, alongside other youth-education topics such as drugs and smoking, helped the project in its early years, but the economic model was not sufficient to sustain the free service provided to students. Mrs. Suttana began building houses to create a homestay that would generate resources to support school activities, building house after house since 2013. The homestay now comprises 10 houses of 2 to 6 occupancy each, for a total capacity of 40 people.

Visitor numbers are roughly 30 to 50 people per visit, with about 10 delegations per year, totaling roughly 300 to 500 visitors annually. Uncontrollable external factors affect the business, such as the flood season, when booked guests cancel and request refunds, which the school must honor. Low season for lodging generally runs from March to May, coinciding with the local haze season, when hot weather discourages tourists.

The Farmer School has air conditioning in only one building. However, the school wants to preserve the authentic character of a 'Farmer School' experience close to nature, showcasing Tai Lue culture. Buildings feature many windows and doors for airflow, natural-material ceilings, and gaps in the walls for ventilation. Installing air conditioning would require full renovation, including replacing all ceilings, which the family is reluctant to do given the cost. As a result, the school closes for four months, March–June, generating no income while monthly operating costs continue regardless; during the 8 income-generating months, they must save enough to cover the 4 closed months, managing the whole year's budget around this cycle.



Competition has increased over time — there are now many more accommodations and more diversified tourism options — and repeat visits are rare, with most tourists who come once not returning, further reducing visitor numbers over time.

Business Model and Networking

Funding sources and financial estimates

In the early period the Farmer School had no external budget; all administrative funding came from Mrs. Suttana's personal savings. Later, Mrs. Jongrak sought some funding to help cover snack/meal costs during weekend art classes, though weekend art teaching remained funded largely from Mrs. Suttana's own pocket. As the school expanded from one to three classrooms, costs rose, prompting the construction of guest accommodation to generate income, with guests sometimes donating toward meals and snacks for the children. Income from the accommodation business mostly went back into administration and ongoing construction of new buildings, since only two guest buildings existed initially and construction costs for new units were high. Approximate funding breakdown: personal funds, 60%; tourist income, 40% (split into 20% for running activities/services and 20% for new construction).